

Hackathon ML Deep Research Report for ATI Prediction From SomaScan Proteomics

Executive summary

This hackathon project is a high-dimensional tabular-omics binary classification task: predict biopsy-confirmed acute tubular injury (ATI) from pre-biopsy blood-based SomaScan proteomics (6,592 anonymized protein features) plus small clinical covariates, using a starter repository scaffold that must be preserved. `filecite{turn0file1}`

The current (authoritative) prompt confirms that the only dataset physically present in the accessible workspace is a single `train.csv` with 426 rows and 6,597 columns, including `ati` (binary), `feature_0001..feature_6592`, and three clinical covariates (`age`, `sex`, `baseline_egfr_23`); it also confirms no missing values and no exact duplicate samples, and that ATI-positive rows have markedly lower baseline eGFR than ATI-negative rows. `filecite{turn0file1}` This immediately implies (a) a strong “easy signal” from eGFR, and (b) a high risk of over-reliance on clinical proxies that can break under cohort shift, especially if the held-out test set is external (as the starter README reportedly claims). `filecite{turn0file1}`

There is a major inconsistency that must shape implementation priorities: the hackathon summary claims ~1,500 training and 500 validation samples, while the local repo snapshot only has 426 training rows. `filecite{turn0file1}` The most robust strategy is to build a leak-resistant evaluation + training harness that scales from 426 rows to 1,500+ rows without changing code paths, and to treat the local `train.csv` as a development subset until the full challenge data distribution and evaluation metric are verified. `filecite{turn0file1}`

Given the regime ($p \approx 6,595+$; n in the hundreds to low thousands), recent tabular benchmarks strongly support gradient-boosted decision tree (GBDT) families as the primary “anchor” models, with carefully designed out-of-fold (OOF) stacking/blending to gain leaderboard points. ¹ For hackathon score maximization, the recommended end-to-end plan is:

- 1) **Build a leakage-resistant CV/OOF toolkit first** (stratified folds; group folds if any patient grouping exists; all preprocessing inside folds; consistent OOF prediction logging). ²
- 2) **Establish a strong GBDT anchor** (XGBoost now; add LightGBM and/or CatBoost if installable), with early stopping and aggressive feature subsampling suited to high- p settings. ³
- 3) **Add diversity models** (elastic-net logistic regression; reduced-feature TabPFN if feasible) and then **blend/stack with OOF predictions** targeting the likely leaderboard metric (log loss or AUROC). ⁴
- 4) **Treat feature selection as a first-class citizen** and do it strictly inside CV; unstable selection is expected in correlated omics, so use stability wrappers (e.g., Stabl-style noise + frequency) or repeated-seed aggregation. ⁵

The remainder of this report provides a concrete, scaffold-compatible implementation plan; validated modeling options grounded in ≤ 4 -year literature; prioritized experiments with expected gain vs cost; decision rules; and a 48–72 hour execution timeline.

Problem framing and requirements

The explicit task is: **predict presence/absence of ATI (binary) from pre-biopsy SomaScan proteomics and clinical covariates**. The challenge rationale is that biopsy is the gold standard but time-consuming and subjective; the goal is to approximate biopsy proxy labels from blood-based features. A prior paper relevant to this phenotype reports substantial proteomic associations with ATI even after adjusting for clinical covariates, suggesting the proteomic panel contains incremental signal beyond kidney function labs. ⁶

This is difficult for three reasons that matter for leaderboard strategy:

- **p>n and correlated feature blocks:** SomaScan proteomics are high-plex and correlated; hundreds to thousands of features may be weakly informative, creating overfitting pressure and seed instability.
- **Clinical proxy dominance (eGFR trap):** the prompt confirms strong separation in eGFR between ATI+ and ATI– in the local dataset, which can produce deceptively high CV if the test distribution differs.
- **Potential cohort shift:** the local README reportedly references an external held-out cohort (KPMP) and ComBat correction, while the hackathon summary describes a different normalization (ANML) and a held-out judge test set. Cohort shift is a primary leaderboard risk for clinical ML, and it strongly influences validation and calibration choices. ⁷

Problem statement

Learn a scoring function $s(x) \in [0, 1]$ where x consists of 6,592 anonymized proteomic features and a few clinical covariates, and $s(x)$ estimates $P(\text{ATI} = 1 \mid x)$.

Constraints (explicit vs unspecified)

- **Confirmed constraints:** must preserve the starter scaffold roles (`preprocess.py`, `model.py`, `evaluate.py`, `train.py`, `predict.py`, `predict.sh`) and treat `Felix/kidney` as canonical.
- **Compute limits:** unspecified; the prompt requests assuming no specific compute limits unless stated.
- **Time limit:** implicitly a hackathon; this report assumes a typical 48–72 hour sprint as requested.
- **Team size:** unspecified.
- **External metadata availability:** explicitly missing locally (no mapping from `feature_XXXX` to protein IDs), which blocks biomarker-prior strategies unless the organizers provide a mapping file later.

Success criteria (score-maximizing)

Success should be defined operationally as:

- **Primary:** maximize the official leaderboard metric on the held-out test set (likely log loss or AUROC; must be verified). `filecite[turn0file1]`
- **Reliability:** improvements must replicate across repeated CV seeds and not depend on leakage-prone feature selection. ⁸
- **Robustness:** combined models should maintain performance when the strongest clinical proxy (`baseline_egfr_23`) is removed or down-weighted (a stress test for cohort shift). `filecite[turn0file1]`
- **Submission safety:** inference must be deterministic, aligned to saved feature order, and tolerant to missing columns in input (predict script already reindexes/fills NaNs). `filecite[turn0file1]`

Dataset description and audit plan

Confirmed local dataset schema and properties

The current prompt confirms the following about the locally present `MedAI/data/data/train.csv`:

- **Rows / columns:** 426 rows, 6,597 columns. `filecite[turn0file1]`
- **Columns:** `sample_id`, `ati` (binary), `feature_0001..feature_6592`, `age`, `sex`, `baseline_egfr_23`. `filecite[turn0file1]`
- **Label balance:** 226 negatives, 200 positives. `filecite[turn0file1]`
- **Missingness / duplicates:** no missing values, `sample_id` unique, no exact duplicate rows, and no duplicate feature rows after numeric checks. `filecite[turn0file1]`
- **Data type hints:** protein values appear already log2-transformed/normalized (global min ~4.0, max ~18.9; typical per-feature std ~0.52). `filecite[turn0file1]`
- **Clinical quirks:** `age` appears binned into decade midpoints; `sex` coded as 1.0/2.0; `baseline_egfr_23` ranges 3–142 and is substantially lower in ATI-positive cases. `filecite[turn0file1]`

What is implied but unconfirmed and must be verified

The prompt also describes an inconsistency between the local README and the hackathon summary:

- Hackathon summary: **1,500 training + 500 validation**, data “log2-normalized and ANML-normalized,” final judged on held-out test. `filecite[turn0file1]`
- Local README: **426 patients**, data “log2-normalized, ComBat-corrected,” external KPMP test cohort, evaluation “log loss.” `filecite[turn0file1]`

ANML is an official SomaScan normalization approach that uses an external reference to adjust variability, documented by the vendor. ⁹ ComBat is a widely used batch correction (not re-explained here), but the key point is the **normalization story differs**, which can change covariate shift behavior and model robustness priorities. `filecite[turn0file1]`

Audit checklist to run immediately when the “real” challenge data arrives

This checklist is designed to protect leaderboard score under distribution shift and prevent false wins caused by leakage.

Data integrity and schema:

- Confirm that proteomic features are consistently named and ordered (`feature_0001..feature_6592` or similar), and that any added covariates are stable across splits. `filecite\turn0file1`
- Confirm whether **validation.csv** (500 rows) exists separately or is only an evaluation service input; do not assume local availability. `filecite\turn0file1`
- Verify label encoding and absence/presence of `ati` in any files expected to be “test.” `filecite\turn0file1`

Leakage and duplicates:

- Hash-based duplicate detection across train/val/test (exact duplicates and near duplicates). In high-p data, near duplicates can occur via repeated samples; if present, group splits are mandatory. ¹⁰
- If any patient ID exists (or can be derived), switch to **StratifiedGroupKFold** to ensure groups appear in exactly one fold’s validation split. ¹¹

Normalization and batch effects:

- Check for plate/batch indicators (even if not labeled); if present, test whether folding stratification inadvertently isolates batches. If batch correction differs between local and full data, expect score instability. `filecite\turn0file1`
- Confirm whether values are already standardized/log2; do not re-normalize unless there is a documented mismatch. (SomaScan standardization steps and ANML are vendor-specified and already intended to reduce technical variation.) ¹²

Clinical proxy risk:

- Quantify baseline performance with **clinical-only** features and specifically with `baseline_egfr_23` alone to estimate how much score may come from a proxy. `filecite\turn0file1`
- Stress test: evaluate models with eGFR removed. If performance collapses, prioritize proteomic signal capture and calibration rather than more tree capacity. `filecite\turn0file1`

Evaluation metrics, validation design, and leakage prevention

Evaluation metrics

The starter evaluation code currently computes foldwise AUROC and log loss, and the starter XGBoost is configured with log loss (`eval_metric=logloss`). `filecite\turn0file1` However, the official judging metric is not definitively confirmed due to the README vs hackathon summary mismatch.

`filecite\turn0file1`

In this uncertainty, the score-maximizing principle is:

- If **log loss** is the leaderboard metric (explicitly mentioned in the local README), then optimize probabilistic accuracy and calibration, not just ranking.
- If **AUROC** is the leaderboard metric, calibration still helps indirectly (stability, stacking), but it is secondary to discrimination.

Because log loss is common in probabilistic clinical competitions and is already implemented in the scaffold, the safest default is to treat **log loss as primary** in model selection, while tracking AUROC as a sanity check.

Leakage prevention rules (non-negotiable)

Leakage is especially dangerous in p>>n: selecting features or scaling using the full dataset before CV can inflate AUC by large margins, and this has been quantified empirically. ⁸ Therefore:

- All transformations (imputation, scaling, feature selection, PCA) must be fit **inside each training fold only**, then applied to that fold's validation data. ¹³
- Any stacking must use **OOF predictions** for the meta-model; training the meta-model on in-fold predictions leads to optimistic bias. scikit-learn's stacking documentation explicitly frames stacking as using cross-validated (OOF) predictions for the final estimator. ¹⁴

Validation strategy recommendation

Given current confirmed data (426 rows, no grouping ID), **StratifiedKFold** with shuffling and multiple repeats is the pragmatic default. ¹⁵ If patient or batch group IDs appear later, switch to **StratifiedGroupKFold** to prevent group leakage. ¹¹

Recommended CV harness (hackathon-practical, leakage-resistant):

- Base: 5 folds (matches starter setting).
- Repeats: 3 random seeds for fold shuffling to estimate variance and stabilize selection (15 fits per model configuration). This is a lightweight alternative to heavy nested CV and is usually feasible in 48–72 hours. ¹⁶
- Primary selection metric: mean log loss across folds; secondary: AUROC and its standard deviation.

Calibration strategy (only if log loss matters)

Many strong discriminative models (especially boosted trees) are not automatically well-calibrated, and calibration is a direct lever for log loss. scikit-learn's calibration tooling supports sigmoid (Platt-style) and isotonic calibration and provides cross-validated calibration workflows designed to keep calibration data disjoint from fit data. ¹⁷

Recommendation for this setting:

- Use **CalibratedClassifierCV** with `method="sigmoid"` as default (more stable on small data than isotonic), calibrated strictly within CV to avoid leakage. ¹⁸

- Apply calibration as a **post-processing step** on top of each base model (or only on the final blended predictor if the pipeline is simpler). ¹⁹

Recent methods and evidence base for this task shape

This survey prioritizes 2022–2026 primary sources and official implementations, focusing on what is most actionable for high-p, medium-n omics tabular classification.

Evidence for GBDT dominance on tabular data

A large benchmark study (NeurIPS 2022) systematically compared tree-based methods and deep tabular models across many datasets and concluded that tree-based models remain state-of-the-art on typical medium-sized tabular problems, with deep models requiring careful tuning and often struggling with uninformative features. ¹⁶ This directly matches the proteomics regime: many features, modest sample sizes, correlated redundancy.

Retrieval-augmented and foundation tabular models as “diversity” add-ons

- **TabR (ICLR 2024)** introduces a retrieval-augmented deep learning model that incorporates k-nearest neighbor-like retrieval into a feedforward network and reports strong results on benchmark suites. ²⁰ Its value in a hackathon is as an ensemble diversity source if a teammate can implement quickly, not as a guaranteed replacement for tuned GBDTs.
- **TabPFN (Nature, published 2025)** presents a tabular foundation model designed for small-data tabular prediction and emphasizes strong out-of-the-box performance up to ~10,000 samples within its operating regime, with support for categorical data and missingness in the updated model. ²¹ Official implementations and documentation are maintained by PriorLabs. ²² Practical constraint: TabPFN and related ICL-style models are commonly framed around feature counts on the order of hundreds; in this proteomics setting it should be used only after aggressive within-fold feature reduction.
- **TabICL (arXiv 2025)** proposes a scalable in-context learning model for larger tables and reports speedups and strong benchmark performance while still discussing typical feature regimes in the hundreds. ²³ It is best treated as research context unless an official package is already available in the environment.

High-dimensional biomarker selection and stability

- **Stabl (Nature Biotechnology 2024)** explicitly targets the “sparse but reliable biomarker set” problem by combining noise injection with data-driven thresholding in multivariable models, motivated by the instability of standard sparse selection under correlation. ²⁴ In hackathon terms: this supports using **stability-aware selection frequency** (repeated subsampling/perturbation) rather than trusting one L1 fit.
- The danger of feature selection leakage is quantified: applying feature selection incorrectly prior to cross-validation can bias AUROC substantially, and the issue is exacerbated in high-dimensional settings. ⁸

Proteomics applications that mirror this modeling regime

A prominent recent proteomics modeling paper (Nature Medicine 2024 proteomic aging clock) provides a real-world template: high-dimensional plasma protein inputs, ensemble-style modeling, and feature reduction for robustness across cohorts. ²⁵ Even though the target is different (age), it is an unusually relevant reference point for “many proteins, generalization across cohorts.”

SomaScan measurement and normalization context

SomaScan is an aptamer-based, high-plex proteomics platform, and vendor technical notes describe standardization and ANML normalization as steps intended to reduce technical variability and improve longitudinal stability. ²⁶ This matters because if the organizers already supply ANML-normalized inputs, additional global scaling may be unnecessary and can sometimes harm signal.

Recommended end-to-end plan and scaffold-compatible implementation design

This section is deliberately implementation-oriented and maps directly onto the existing file responsibilities without proposing a new framework. `filecite{turn0file1}`

End-to-end pipeline design

```
flowchart LR
    A[Raw CSV: train/val/test] --> B[preprocess.py<br/>schema checks + feature assembly]
    B --> C[evaluate.py<br/>splitter + fold pipeline + OOF store]
    C --> D[model.py<br/>model zoo + configs]
    D --> E[OOF predictions for each base model]
    E --> F[Ensemble layer<br/>blend or stack]
    F --> G[Optional calibration<br/>sigmoid or isotonic]
    G --> H[train.py<br/>fit full-data final models + save weights]
    H --> I[predict.py<br/>load weights + align feature order + infer]
```

This mirrors the starter scaffold’s separation of responsibilities while enabling feature ladders, calibration, and stacking without leakage. `filecite{turn0file1}` ²⁷

Feature strategy: a ladder that works with anonymous features

Because `feature_XXXX` are anonymized and there is no mapping file locally, all “biomarker prior” strategies must be conditional: only activate if a mapping appears later. `filecite{turn0file1}` The ladder below is designed to work now and upgrade later.

FS0 (clinical-only): `age`, `sex`, `baseline_egfr_23`.

FS1 (all features): all `feature_` columns + clinical.

FS2 (within-fold filtered top-k): all features ranked inside each fold (by univariate AUC, mutual

information, or model importance), keep top $k \in \{256, 512, 1024, 2048\}$.

FS3 (stability-selected set): repeated within-fold selection frequency (Stabl-inspired), keep features exceeding a frequency threshold, then train final models on that reduced set. ²⁸

If an organizer-provided mapping from `feature_XXXX` → protein name appears, add an additional rung:

FS_BKBC (literature prior): FS0 + proteins reported associated with ATI in the relevant cohort paper, used as an evidence-based reduced panel. ⁶

(The older provided PDF emphasizes this conceptually, but it cannot be implemented without a mapping file in the current snapshot.)

Model architectures and training regimens

The environment is confirmed to have XGBoost and scikit-learn, but not LightGBM or TabPFN by default. ^{filecite[turn0file1]} The plan therefore prioritizes “works now” baselines, with optional installs for higher upside.

Primary (anchor) model: XGBoost (works in current environment)

Use the official XGBoost implementation with early stopping; high-p requires strong feature subsampling and regularization. ²⁹

Suggested baseline hyperparameter templates (to sweep lightly, not exhaustively):

- **XGB-A (conservative, robust):**
 - `objective` : `binary:logistic`
 - `eval_metric` : `logloss` (track AUC too)
 - `tree_method` : `hist` (CPU)
 - `n_estimators` : 4000 (with early stopping 200)
 - `learning_rate` : 0.02–0.05 (start 0.03)
 - `max_depth` : 3–5 (start 4)
 - `min_child_weight` : 5–20 (start 10)
 - `subsample` : 0.7–0.9 (start 0.8)
 - `colsample_bytree` : 0.10–0.30 (start 0.20)
 - `reg_lambda` : 3–15 (start 8)
 - `reg_alpha` : 0–1 (start 0.1)
- **XGB-B (more capacity, only if stable):**
 - `max_depth` : 6
 - `colsample_bytree` : 0.15
 - stronger L2 if overfitting

Training regimen:

- Loss: log loss (native), optimize early stopping within each fold. ³⁰

- Seeds: run 3–5 seeds and average predictions (cheap variance reduction).
- Feature set: start FS1, then FS2 if stability improves log loss.

Optional GBDT upgrades: LightGBM and CatBoost

If you can install them, these are high-ROI additions because they increase ensemble diversity while staying in the GBDT family.

- LightGBM official code and docs are actively maintained (repository moved in March 2026). ³¹
- CatBoost official code supports CPU/GPU and is often strong with categorical covariates (like `sex`) without heavy preprocessing. ³²

LightGBM template (if available):

- `objective` : `binary`
- `metric` : `binary_logloss` (plus AUC)
- `n_estimators` : 5000 with early stopping 150
- `learning_rate` : 0.02–0.05
- `num_leaves` : 31–127 (start 63)
- `min_data_in_leaf` : 20–80 (start 50)
- `feature_fraction` : 0.15–0.35
- `bagging_fraction` : 0.7–0.9 with `bagging_freq=1`
- `lambda_l2` : 3–15, `lambda_l1` : 0–1
- Handle missing values with built-in missing support if full data later contains NaNs. ³³

CatBoost template (if available):

- Use GPU if permitted; otherwise CPU. ³²
- Treat `sex` as categorical explicitly.

Diversity model: Elastic-net logistic regression (fast, calibration-friendly)

The starter includes an L1 logistic regression with scaling. `filecite\turn0file1` Upgrade it to elastic net and treat it as both (a) a strong linear baseline in $p \gg n$ and (b) a feature-screening engine, while addressing instability.

Training regimen:

- Scale features inside fold (Pipeline).
- Use `saga` solver to support elastic net; sweep `l1_ratio ∈ {0.1, 0.5, 0.9}`, `C` logarithmic grid.
- Use class weights if imbalance increases in the full dataset.
- Stability wrapper (Stabl-inspired): subsample rows (e.g., 70–90% of train fold) multiple times, tally selected features, and keep those exceeding a frequency threshold. ³⁴

Reduced-feature foundation model (optional): TabPFN

TabPFN is compelling for small-data regimes and has an official implementation and docs, but it must be used after reduction because proteomics p is far beyond its typical “hundreds of features” operating regime. ³⁵

Practical hackathon role:

- Only attempt after you have a stable GBDT baseline and a reliable feature reduction method (FS2/FS3 producing ≤ 500 features within-fold).
- Use as a diversity model for blending/stacking, not as the sole solution. ³⁶

Ensembling and postprocessing plan (score-focused)

- 1) **Seed averaging (almost free):** average predicted probabilities from 3–5 seeds for each GBDT model.
- 2) **Simple blending:** convex combination of base model probabilities; tune blend weights on OOF predictions to minimize log loss (or maximize AUC depending on verified metric).
- 3) **Stacking (highest upside if implemented correctly):** use OOF predictions as features and train a regularized meta-learner (ridge logistic regression). scikit-learn’s stacking formulation provides a reference for the correct OOF-based approach. ³⁷
- 4) **Calibration:** calibrate either each base model or the final blended predictor with `CalibratedClassifierCV` (sigmoid) if log loss is the target. ¹⁸

Postprocessing:

- Never hard-threshold for submission unless the competition explicitly demands class labels; log loss and AUROC both prefer continuous probabilities. The scaffold’s 0.5 threshold should remain only for diagnostics. `filecite\turn0file1`

Implementation notes mapped to each scaffold file

preprocess.py (shared data loading / feature assembly) `filecite\turn0file1`

Add:

- A strict schema validator: checks presence of `sample_id`, protein prefix columns, and clinical covariates if expected.
- Feature-set selection mechanism (FS0/FS1/FS2/FS3) driven by config; FS2/FS3 should accept a fold-specific feature list produced in `evaluate.py`.
- Option to treat `sex` as categorical (keep numeric for XGB; mark categorical index for CatBoost when used).

model.py (model definitions and constants) `filecite\turn0file1`

Add:

- A model registry that returns a sklearn-compatible estimator or Pipeline per model name: `xgb_anchor`, `lgbm_anchor`, `catboost_anchor`, `elastic_net`, `tabpfn_reduced`.
- Config dataclasses (or dicts) containing hyperparameter “templates” and a small sweep list to control compute.

- A shared `get_splitter()` that returns `StratifiedKFold` or `StratifiedGroupKFold` depending on whether groups exist (groups default `None`). ³⁸

evaluate.py (CV experimentation entry point) `filecite\turn0file1`

Upgrade to:

- A fold loop that (a) stores OOF preds per model per seed, (b) fits any feature selection strictly inside folds, and (c) logs mean/std of log loss and AUC. ¹³
- A clean OOF artifact format (e.g., parquet/csv): `sample_id`, `y_true`, `fold`, `seed`, `pred_model_X`.
- An ensemble evaluator that takes OOF matrices and computes blend weights and/or trains a stacked meta-model.

train.py (final train on all training data and save weights) `filecite\turn0file1`

Add:

- A “final recipe” that trains the chosen base models on all training data using the best-validated configs, plus their selected feature sets (e.g., FS3 stable features).
- Save: model weights per base model, calibration objects (if used), meta-model (if stacking used), and the definitive feature column ordering used for prediction.
- Ensure any feature selection used to define FS3 for final training is produced from training data only (no validation/test leakage).

predict.py (inference entry point) `filecite\turn0file1`

Extend to:

- Load multiple base models and (optionally) a meta-model; compute base probabilities and combine them via saved blend weights / stacker.
- Apply saved calibration step (if used).
- Continue to enforce feature alignment by reindexing to the saved feature order and filling missing with NaN (already present behavior). `filecite\turn0file1`

predict.sh (shell wrapper) `filecite\turn0file1`

Only adjust if new dependencies are introduced (LightGBM/CatBoost/TabPFN), ensuring deterministic environment activation.

Reproducibility and “hackathon-safe” engineering practices

- Fix random seeds at three levels: numpy, Python hash seed, and model seeds; still do multi-seed evaluation because $p \gg n$ is inherently stochastic. ¹⁶
- Save every artifact needed for inference: feature order, model(s), and calibration/ensemble parameters; never recompute feature selection at inference time. `filecite\turn0file1`
- Keep a single source of truth for folds (store fold assignment by `sample_id`), so OOF predictions from different models are perfectly aligned.

Prioritized experiments, decision rules, timeline, and open questions

Candidate-model comparison table

Candidate	Role in final solution	Why it fits this task	Key risks	Expected hackathon value
XGBoost (GBDT)	Primary anchor	Strong tabular baseline; official, scalable; works now in environment ²⁹	Can overfit in high-p if colsample too high; calibration often needed for log loss ¹⁹	Very high
LightGBM (GBDT)	Anchor + diversity	Efficient GBDT; official repo/docs; good with feature subsampling ³⁹	Not installed by default; install friction <code>!pip install lightgbm</code>	High if installable
CatBoost (GBDT)	Diversity GBDT	Handles categorical covariates; strong tabular performance; official GPU support ³²	Install + runtime; tuning overhead	Medium-high
Elastic-net logistic regression	Diversity + selection	Stable linear baseline, often strong in omics; calibration-friendly; fast ⁴⁰	Feature selection instability without stability wrapper ²⁸	High
TabPFN (foundation model)	Reduced-feature diversity	Strong small-data results; official implementation/docs ³⁵	Needs aggressive reduction (≤ 500 feats); may be hard to integrate fast ⁴¹	Medium (high upside if smooth)
TabR (retrieval DL)	Optional diversity	Strong deep tabular results; official implementation exists ⁴²	Engineering complexity; time-to-iterate	Medium (team-dependent)

Augmentation and regularization strategies table

In tabular proteomics, “augmentation” should be treated as regularization rather than image-style transforms.

Strategy	Where to apply	Expected benefit	Compute cost	Notes
Row subsampling / bagging (GBDT)	GBDTs	Variance reduction; robustness	Low–medium	Use subsample + multi-seed averaging
Feature subsampling (colsample/feature_fraction)	GBDTs	Critical in high-p to prevent spurious splits	Low	Primary lever for $p \gg n$ stability
Noise injection (Stabl-inspired)	Linear selection (FS3)	Stabilizes selected features under correlation	Medium	Supported by Stabl's design rationale ⁴³
Label smoothing / mixup	Usually skip	Unclear in binary clinical labels	—	Risky unless labels are noisy and metric is log loss
Synthetic tabular generation	Optional	Can help imbalance, but risky	High	Only if time remains after strong baselines

Ensembling strategies table

Method	Implementation complexity	Expected gain	Leakage risk	Recommended usage
Seed averaging	Low	Small but consistent	Low	Do for every GBDT
Simple blending of base probs	Low	Moderate	Low	First ensemble stage
Stacking (OOF meta-model)	Medium	Moderate–high	Medium if mis-implemented	Use scikit OOF-aligned approach ³⁷
Calibration of final predictor	Medium	High if log loss metric	Medium if data not disjoint	Use CalibratedClassifierCV correctly ⁴⁴
AutoGluon Tabular stacking/bagging	Medium–high	Potentially high	Low if used as intended	Consider if allowed; it builds ensembles/bagging internally ⁴⁵

Prioritized experiment list with estimated impact and cost

Estimates assume tabular CPU training on hundreds–low-thousands of rows; actual timing depends on hardware and whether LightGBM/CatBoost are used.

1) Leakage-resistant CV + OOF logging (foundation work)

- Expected impact: enables every later gain; prevents false progress.
- Cost: 2–4 engineering hours.
- Success: reproducible OOF metrics across 3 seeds; no preprocessing outside folds. 13

2) XGBoost tuned anchor on FS1 with early stopping + multi-seed averaging

- Expected impact: large.
- Cost: 2–6 hours; compute moderate.
- Success: improves over starter XGB (100 trees) and reduces log loss meaningfully. 46

3) Clinical-only vs all-features ablation + eGFR removal stress test

- Expected impact: directs the whole strategy (proxy risk).
- Cost: 1–2 hours.
- Success: combined model materially improves over eGFR-only; performance does not collapse when eGFR removed. filecite turn0file1

4) Elastic-net LR as a diversity model (properly tuned)

- Expected impact: moderate; often improves ensembles and calibration.
- Cost: 1–3 hours.
- Success: competitive log loss on its own; improves blended log loss.

5) Within-fold top-k feature filtering (FS2) feeding XGB and LR

- Expected impact: moderate to high if full dataset is noisy.
- Cost: 2–5 hours.
- Success: lower variance across seeds; improved log loss.
- Guardrail: must be fold-internal to avoid bias. 8

6) Blend weights optimized on OOF predictions

- Expected impact: moderate; often “free leaderboard points.”
- Cost: <1 hour.
- Success: blended OOF log loss improves vs best single model.

7) Stacking with ridge logistic meta-model + optional calibration

- Expected impact: moderate–high if base models are diverse.
- Cost: 2–6 hours.
- Success: consistent improvement across repeated CV seeds; no leakage. 47

8) Optional installs and additions: LightGBM/CatBoost

- Expected impact: moderate; increases ensemble diversity.
- Cost: 1–3 hours install + tuning.

- Success: either beats XGB or adds complementary errors for stacking. 48

9) Stability-selected FS3 (Stabl-inspired frequency selection) feeding final GBDT

- Expected impact: moderate; especially helpful under shift.
- Cost: 4–10 hours.
- Success: reduced feature set leads to improved and more stable OOF metrics. 28

10) TabPFN on ≤ 500 features as an ensemble member (only if smooth)

- Expected impact: uncertain; can be high if it generalizes differently.
- Cost: 2–6 hours integration + reduction.
- Success: adds measurable gain in blend/stack without destabilizing. 49

Expected gain vs cost chart

```
quadrantChart
title Expected leaderboard gain vs engineering/compute cost
x-axis Low cost --> High cost
y-axis Low gain --> High gain
quadrant-1 High gain / High cost
quadrant-2 High gain / Low cost
quadrant-3 Low gain / Low cost
quadrant-4 Low gain / High cost
"Leak-safe CV + OOF logging": [0.15, 0.95]
"XGB tuned + seed averaging": [0.30, 0.85]
"Elastic-net baseline": [0.25, 0.55]
"OOF blend weights": [0.15, 0.60]
"Stacking + calibration": [0.55, 0.75]
"FS2 top-k filtering": [0.50, 0.65]
"LightGBM add-on": [0.55, 0.60]
"Stability FS3": [0.75, 0.55]
"TabPFN reduced-features": [0.70, 0.50]
"TabR integration": [0.85, 0.35]
```

Actionable 48–72 hour timeline and deliverables

```
gantt
title 48–72 hour hackathon execution plan
dateFormat HH
axisFormat %Hh

section Foundation
Repo audit + schema validator + fold harness      :a1, 00, 4
OOF prediction storage + metric reporting          :a2, 04, 3
```

section Strong baselines

XGB anchor tuning (FS1) + multi-seed averaging	:b1, 07, 8
Clinical-only vs all-features + eGFR stress tests	:b2, 10, 4
Elastic-net sweep (proper CV pipelines)	:b3, 14, 5

section Feature reduction + ensembles

FS2 within-fold top-k filtering + retrain XGB/LR	:c1, 19, 8
OOB blending + weight tuning	:c2, 24, 3
Stacking meta-model + validation	:c3, 27, 6

section Finalization

Calibration (if log loss) + reliability checks	:d1, 33, 5
Train full-data final models + save artifacts	:d2, 38, 3
Submission rehearsal + inference sanity checks	:d3, 41, 4

Key deliverables (by the end):

- A reproducible CV report artifact: mean/std log loss and AUROC over repeated seeds, plus OOF prediction tables for each model. ⁵⁰
- Saved final model bundle: base model weights, ensemble weights/meta-model, calibration objects (if used), and feature column ordering for inference. `filecite{turn0file1}`
- A one-page “submission recipe” describing exactly which config produced the submitted predictions, for auditability.

Operational decision rules (if/then)

These rules are designed to prevent wasted time and optimize leaderboard score under uncertainty.

- If **log loss is confirmed as the judging metric**, then prioritize calibration and ensembles even if AUROC does not improve, because calibrated probabilities directly affect log loss. ¹⁹
- If **LightGBM or CatBoost cannot be installed quickly**, do not stall: proceed with XGBoost + elastic net + stacking; ensemble diversity matters more than chasing an alternative GBDT at the cost of iterations. `filecite{turn0file1}`
- If **FS2/FS3 feature selection changes drastically across folds/seeds**, treat that as instability; fall back to FS1 with strong colsample regularization and seed averaging, or use stability thresholds (selection frequency). ²⁸
- If **clinical-only performance is nearly as good as all-features**, expect shift risk; keep the proteomics model as a separate ensemble member and emphasize robust feature reduction to find incremental proteomic signal. `filecite{turn0file1}`
- If **stacking improves OOB but degrades on the provided validation split**, suspect leakage or overfitting; revert to weighted blending and add calibration. ⁵¹
- If a **feature mapping file** (feature→protein) becomes available, then activate the literature-prior rung (FS_BKBC) and compare it to FS2/FS3; otherwise, keep everything anonymous and importance-based. `filecite{turn0file1}` ⁶

Explicit assumptions made for unspecified items

- The official leaderboard accepts probabilistic predictions and does not require hard labels (common for log loss / AUROC competitions).
- There is no patient-level grouping in the delivered data unless a group column appears later; if it appears, group CV will be used. ¹¹
- Compute is “reasonable hackathon compute” (multi-core CPU, optional single GPU), consistent with the request to assume no strict compute limit. `filecite\turn0file1`
- The local `train.csv` may be a subset or dev snapshot; the final data distribution and the presence of a separate validation file may differ. `filecite\turn0file1`

Open questions and verification tasks ranked by importance

The following unknowns should be resolved before committing to the final modeling recipe:

- **Official judging metric (log loss vs AUROC vs other):** the strategy for calibration and ensemble selection depends on this. `filecite\turn0file1`
- **True data inventory and sizes:** reconcile the 426-row local snapshot with the claimed 1,500/500 split; confirm what files are provided at runtime. `filecite\turn0file1`
- **Train/test cohort relationship:** confirm whether the test data is an external cohort (KPMP mentioned in README) or a random split; this changes how aggressively you should regularize and how much to trust clinical proxies. `filecite\turn0file1`
- **Feature mapping availability:** if `feature_XXXX` → protein mapping exists, literature-prior panels become implementable; otherwise, keep anonymous pipelines. `filecite\turn0file1`
- **Normalization specifics (ANML vs ComBat vs other):** confirm which processing has already occurred and whether batch indicators exist; avoid double-normalization. `filecite\turn0file1` ⁵²

¹ ¹⁶ Why do tree-based models still outperform deep learning on tabular data?

https://arxiv.org/abs/2207.08815?utm_source=chatgpt.com

² ¹³ 11. Common pitfalls and recommended practices

https://scikit-learn.org/stable/common_pitfalls.html?utm_source=chatgpt.com

³ ²⁹ dmlc/xgboost: Scalable, Portable and Distributed Gradient ...

https://github.com/dmlc/xgboost?utm_source=chatgpt.com

⁴ ¹⁷ ¹⁸ CalibratedClassifierCV

https://scikit-learn.org/stable/modules/generated/sklearn.calibration.CalibratedClassifierCV.html?utm_source=chatgpt.com

⁵ ²⁴ ²⁸ ³⁴ ⁴³ Discovery of sparse, reliable omic biomarkers with Stabl

https://www.nature.com/articles/s41587-023-02033-x?utm_source=chatgpt.com

⁶ Plasma proteomics of acute tubular injury | Nature Communications

https://www.nature.com/articles/s41467-024-51304-x?utm_source=chatgpt.com

⁷ ¹⁹ ⁴⁰ ⁴⁴ 1.16. Probability calibration

https://scikit-learn.org/stable/modules/calibration.html?utm_source=chatgpt.com

⁸ ⁵⁰ Measuring the bias of incorrect application of feature selection ...

https://pmc.ncbi.nlm.nih.gov/articles/PMC8613324/?utm_source=chatgpt.com

9 12 52 **SomaScan® v4.0 and v4.1 Data Standardization**

https://somalogic.com/wp-content/uploads/2022/07/SL00000442_Rev4_2021-07_-SomaScan-v4.0-and-v4.1-Data-Standardization.pdf?utm_source=chatgpt.com

10 11 38 **StratifiedGroupKFold — scikit-learn 1.8.0 documentation**

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.StratifiedGroupKFold.html?utm_source=chatgpt.com

14 27 37 47 51 **StackingClassifier**

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https://openreview.net/forum?id=rhgIgTSSxW&utm_source=chatgpt.com

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https://github.com/microsoft/LightGBM/blob/master/docs/Parameters.rst?utm_source=chatgpt.com

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https://openreview.net/pdf?id=rhgIgTSSxW&utm_source=chatgpt.com

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https://auto.gluon.ai/dev/api/autogluon.tabular.models.html?utm_source=chatgpt.com